BIZSAGE

Restaurant Operations Expert System

Project Report

A rule-based decision-support application for investigating restaurant performance and comparing practical actions.

Project type	Knowledge-based expert system with an integrated decision support system
Application	Python and Streamlit web application
Domain	Financial and operational decision support for small restaurants
Live application	https://bizsage.streamlit.app/

Educational project. Demo businesses and branch records are fictional.

Table of contents

1. Introduction 3

1.1 Project objectives . . . 3

1.2 Scope . . . 3

2. System explanation 3

2.1 End-to-end workflow . . . 4

2.2 Inference and certainty . . . 4

2.3 Decision-support method . . . 4

3. Market use cases 4

3.1 Independent restaurant review . . . 4

3.2 Multi-branch and franchise review . . . 5

3.3 Highway and high-variation staffing . . . 5

3.4 Educational and advisory use . . . 5

4. User interface and screenshots 5

4.1 Start page . . . 6

4.2 Business information page . . . 7

4.3 Investigation page . . . 8

4.4 Findings and options page . . . 9

4.5 Branch comparison . . . 10

5. Design considerations and limitations 11

6. Conclusion 11

References and project links 11

1. Introduction

Small restaurants make frequent decisions about staffing, prices, purchasing, cash flow, and whether to continue operating a branch. Owners may have useful sales and expense records but limited time or specialist support to connect those figures to likely operational causes. BizSage is designed as an educational decision aid that organizes available information and presents understandable evidence.

The application focuses on Pakistani restaurant examples and everyday business inputs. It accepts incomplete information and uses rule-based reasoning to investigate several possible causes rather than returning a single unexplained answer. It then compares selected actions and leaves the final decision with the owner.

1.1 Project objectives

- Represent restaurant-operation knowledge in an editable declarative rule base.
- Infer and explain possible causes from financial history and answers to follow-up questions.
- Handle unknown, approximate, incomplete, conflicting, and overlapping evidence explicitly.
- Provide a usable Streamlit interface for non-technical users.
- Compare practical alternatives while keeping the human decision-maker in control.

1.2 Scope

BizSage covers six diagnostic categories: revenue and demand; expenses and profitability; cash flow and liquidity; inventory and wastage; staffing and operations; and debt and stability. It also offers decision comparisons for staffing, cost and margin improvement, cash-flow recovery, and branch or franchise decisions.

2. System explanation

BizSage is an expert system because domain knowledge is stored separately from the inference procedure. The JSON knowledge base defines facts, thresholds, hypotheses, conditions, and conclusions. A generic Python engine evaluates those conditions against working memory. The resulting evidence is combined into certainty-factor support, explained to the user, and used to select useful follow-up questions.

Knowledge base	76 declarative rules: 60 unique diagnostic conditions, 14 exclusion rules, and 2 chained rules.
Coverage	Six diagnostic categories, with 10 unique diagnostic conditions per category; 13 possible causes (hypotheses).
Reasoning	Forward chaining: rules that assert new facts can enable additional rules in a later inference cycle.
Decision support	Weighted option scores, criteria trade-offs, pros and cons, and illustrative estimates; the system does not decide for the owner.

2.1 End-to-end workflow

Start: choose a fictional example or begin a new assessment. **Business information:** provide the restaurant profile, monthly history, and current figures. **Investigation:** answer relevant questions, with “not sure” available. **Findings and options:** review evidence and compare possible responses.

The calculation layer derives measurable facts such as revenue changes, margins, cash buffers, and staffing ratios. The inference engine interprets JSON rules using three-valued logic: TRUE, FALSE, or UNKNOWN. A missing fact remains UNKNOWN, so the system does not mistake missing information for a negative response or a zero value.

2.2 Inference and certainty

When all conditions of a rule are satisfied, that rule fires and contributes supporting or opposing evidence to one or more hypotheses. Evidence carries certainty based on its source and answer quality. The hypothesis manager combines positive evidence, negative evidence, and related evidence groups. Related signals are partially down-weighted to reduce double counting.

The question selector prioritizes facts that could clarify unresolved hypotheses. If a rule still needs an unknown fact, the system may ask about it; if answers conflict with calculated or other entered facts, the affected reasoning is paused for clarification. Users can inspect the evidence, missing information, data-quality messages, and rule trace.

2.3 Decision-support method

The DSS evaluates alternatives against configurable criteria such as financial impact, risk, cost, feasibility, and sustainability. Rule-based modifiers adjust an option when the current findings support the adjustment. Scores are normalized and displayed with advantages, disadvantages, rationale, and estimates where required inputs are available.

The ranking is guidance, not an automatic decision. Estimate assumptions are visible and editable; they are illustrative values and are not forecasts. Owners should validate important decisions with appropriate business or accounting professionals.

3. Market use cases

BizSage is a prototype for small and medium-sized restaurant operators who need a structured way to think through operating issues. It may also be used as a teaching or discussion aid by business students, advisers, and instructors. The following are possible use cases, not claims of customer adoption or measured commercial impact.

3.1 Independent restaurant review

An owner enters monthly sales, expenses, food costs, customer counts, and selected current operating figures. BizSage can highlight evidence consistent with declining demand, rising input costs, excessive wastage, cash pressure, or staffing that does not match weekday and weekend demand. The owner can then review which additional information might change the conclusion.

3.2 Multi-branch and franchise review

A manager compares revenue and expenses across branches over time. BizSage summarizes branch margins and trends, checks whether branches move together, and can compare a branch with a selected reference branch. The system then presents options such as keeping a branch open, improving it, or changing its operating model, with the human decision-maker responsible for the final choice.

3.3 Highway and high-variation staffing

A highway restaurant may face different customer loads on weekdays and weekends. By entering customer volumes, staff counts, and wage assumptions, a manager can compare staffing arrangements and see the assumptions behind estimated effects. This supports a discussion about flexible daily-wage staffing without treating the estimate as a guaranteed saving.

3.4 Educational and advisory use

In a classroom or advisory session, the rule trace and evidence explanations can be used to demonstrate knowledge representation, forward chaining, uncertainty, data quality, and decision-support scoring. Fictional demo data makes the workflow reproducible without presenting made-up records as real business evidence.

4. User interface and screenshots

The live application is available at https://bizsage.streamlit.app/. The following pages are intentionally left as screenshot placeholders. Insert your own captures from the deployed or local application before submitting the report.

4.1 Start page

Capture the welcome screen showing sample restaurant choices and the path to enter a business.

Paste or insert your screenshot in this area

4.2 Business information page

Capture the restaurant profile, monthly history, or current-month figures form.

Paste or insert your screenshot in this area

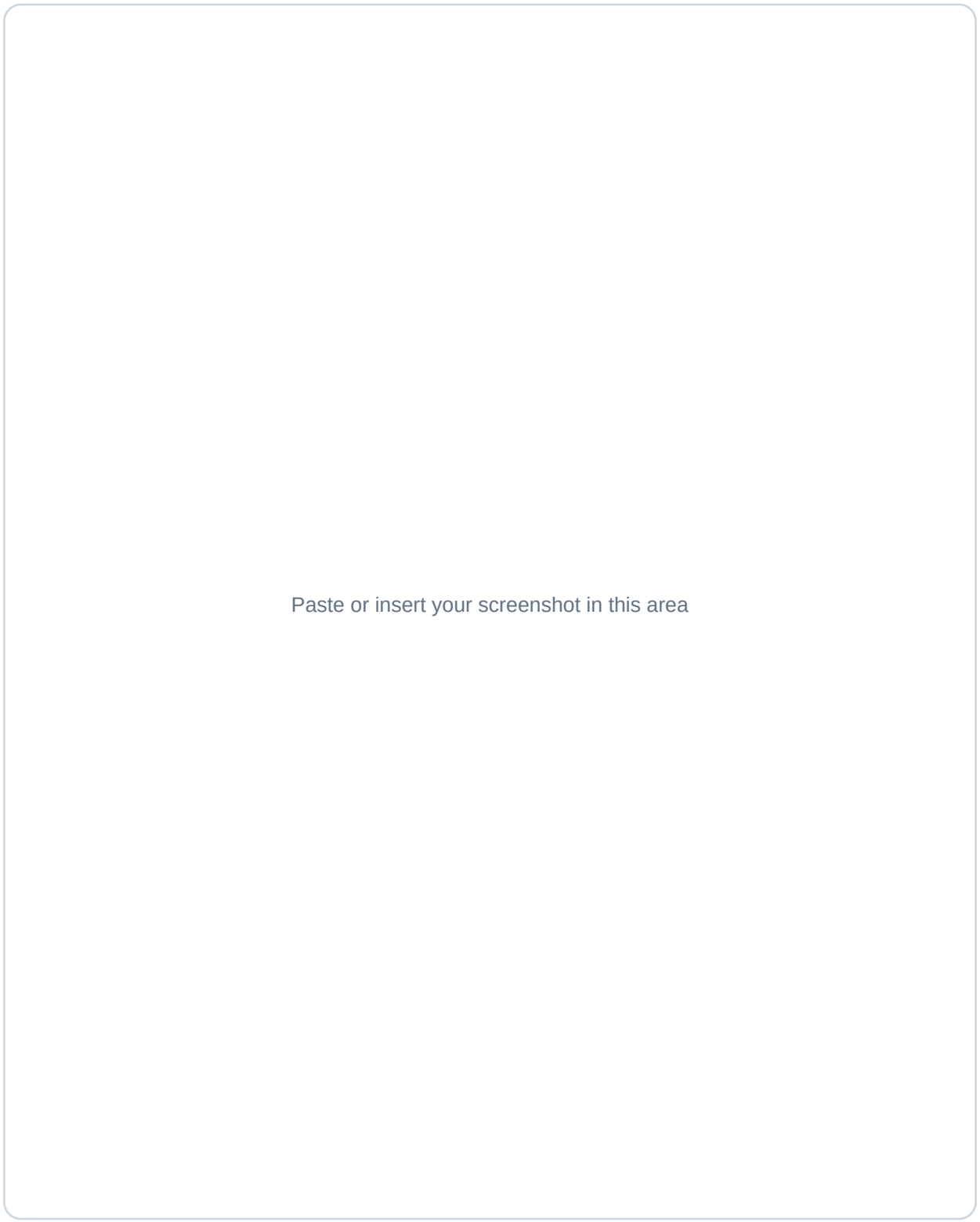
4.3 Investigation page

Capture an example question alongside the live reasoning panel or investigation progress.

Paste or insert your screenshot in this area

4.4 Findings and options page

Capture the supported findings, evidence explanations, and one decision-option comparison.



4.5 Branch comparison

Capture the branch trend/comparison view, including the sample chain or your entered example.

Paste or insert your screenshot in this area

5. Design considerations and limitations

- **Expert-assigned thresholds:** rule thresholds and certainty factors are educational starting values and have not been statistically validated as industry standards.
- **Not probabilities:** support scores represent evidence strength under the defined rules; they are not calibrated statistical probabilities.
- **Illustrative estimates:** option estimates rely on entered values and assumptions and should not be read as predictions or guaranteed savings.
- **Data quality:** incomplete or inconsistent inputs can limit conclusions; review warnings and missing-information prompts.
- **Human responsibility:** BizSage is a decision-support tool, not a substitute for professional financial or business advice.
- **Deployment:** a public Streamlit URL is available, but network performance on mobile data or 3G must be tested separately.

6. Conclusion

BizSage combines a declarative restaurant knowledge base, a Python forward-chaining inference engine, uncertainty-aware hypothesis evaluation, and a weighted decision-support workflow. It aims to make the reasoning visible: users can see what the system found, why it found it, what is still unknown, and how practical alternatives compare. The owner remains responsible for choosing and validating the action.

References and project links

Live application: https://bizsage.streamlit.app/

Source code and configuration: Python modules in `expert_system/`, `decision_support/`, `calculations/`, and `ui/`; declarative knowledge in `knowledge_base/`.

Project tests: `python -m pytest -q`.